

AR PHYSIOTHERAPY COACH

Hackathon Project Proposal — VISIONARY Hackathon 2.0

Domain: Healthcare & Wellness | Platform: WebXR (Browser-based) | Team: 4+ members

1. Problem Statement

Physiotherapy is one of the most critical yet underutilized forms of healthcare in India. Patients recovering from injuries, surgeries, or chronic conditions need consistent supervised exercise — but access is far from guaranteed.

80% of patients	skip follow-up physiotherapy sessions in India due to cost, travel distance, or time constraints — leading to incorrect self-exercise, slower recovery, and frequent reinjury.
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Without a trained eye to watch them, patients unknowingly perform exercises with wrong posture. There is currently no affordable, accessible, browser-based solution that gives real-time visual correction feedback at home.

2. Our Solution

AR Physiotherapy Coach is a browser-based Augmented Reality web app that uses your phone camera to:

- Detect your body pose in real time using MediaPipe Pose (33 keypoints)
- Overlay a colour-coded AR skeleton on your live camera feed via WebXR + Three.js
- Calculate joint angles and compare them against ideal exercise benchmarks
- Give instant visual (green = correct / red = incorrect) and voice feedback
- Track your sessions, rep counts, and improvement over time on a dashboard

No app download. No headset. Just open the link on any smartphone and start your session.

3. Key Features

Real-time Pose Detection	MediaPipe tracks 33 body keypoints at 30fps directly in the browser with no server round-trip needed.
AR Skeleton Overlay	Three.js draws a live skeleton over the camera feed. Bones turn green when posture is correct

	and red when it needs correction.
Joint Angle Analysis	Calculates angles at knee, hip, shoulder, and elbow joints and compares them against physiotherapist-defined ideal ranges for each exercise.
Voice + Visual Feedback	Web Speech API reads out cues like 'Straighten your back' or 'Bend your knee more' in real time.
Exercise Library	Preloaded with common physiotherapy exercises: squats, leg raises, shoulder rotations, neck stretches, and more.
Progress Dashboard	Patients and doctors can view session history, accuracy scores, rep counts, and recovery trends over time.
Doctor Portal	Doctors can remotely prescribe exercise plans, set rep targets, and monitor patient progress asynchronously.

4. Technology Stack

AR & 3D Rendering

- WebXR Device API — browser AR session management
- Three.js — 3D skeleton overlay rendering
- Canvas API — pose landmark drawing

Pose Detection & AI

- MediaPipe Pose — real-time 33-keypoint detection
- Custom angle calculation logic (JavaScript)
- Web Speech API — voice feedback

Frontend

- React.js — UI components and state management
- TailwindCSS — styling
- Vite — build tooling

Backend & Data

- Node.js + Express — REST API
- Firebase Firestore — session and user data
- Firebase Auth — doctor/patient login

5. Team Roles & Responsibilities

Member	Role	Responsibilities
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Person 1	<i>AR & WebXR Lead</i>	Set up WebXR session, Three.js skeleton rendering, camera feed integration, and AR overlay pipeline.
Person 2	<i>Pose Detection Lead</i>	Integrate MediaPipe Pose, write joint angle calculation logic, define correct/incorrect posture thresholds per exercise.
Person 3	<i>Frontend & UX Lead</i>	Design and build patient-facing UI screens, exercise selection, rep counter, voice feedback integration, and mobile responsiveness.
Person 4	<i>Backend & Dashboard Lead</i>	Build Node.js API, Firebase Firestore integration, session tracking, doctor portal, and progress dashboard.

6. Hackathon Build Timeline

Phase 1 (Hours 1–4) — Setup & Foundations

Repo setup, project scaffold (React + Vite), Firebase config, WebXR boilerplate, MediaPipe integration test.

Phase 2 (Hours 5–10) — Core AR Engine

Live camera feed + pose detection running. Skeleton overlay on Three.js. Basic joint angle calculation for 2–3 exercises.

Phase 3 (Hours 11–16) — Feedback & UI

Green/red colour feedback on skeleton. Voice cues via Web Speech API. Patient-facing exercise UI, rep counter.

Phase 4 (Hours 17–20) — Backend & Dashboard

Firebase session storage. Progress dashboard with charts. Doctor portal with exercise prescription.

Phase 5 (Hours 21–24) — Polish & Demo Prep

Mobile optimization, bug fixes, fallback mode (2D pose only), demo script rehearsal, pitch preparation.

7. Demo Plan for Judges

The live demo is our biggest weapon. Here is the exact sequence:

- Open the app in a mobile browser — no download, no setup
- Select 'Squat' from the exercise library
- Point the phone at the person doing the demo
- Perform a squat with wrong posture → skeleton turns red, voice says 'Bend your knees more'

- Correct the posture → skeleton turns green, rep count increases
- Open the progress dashboard to show session history and improvement graphs
- Show the doctor portal — prescribe an exercise plan remotely

Fallback plan: If WebXR is unstable on demo hardware, fall back to 2D pose detection on a plain camera feed. The posture feedback logic remains the same and the impact is identical.

8. Why This Project Wins

Innovation	AR pose correction in a browser — no app needed. Rarely done at this level of polish in a hackathon.
Real-world Impact	Directly addresses a massive healthcare access gap in rural and semi-urban India.
Technical Depth	WebXR + MediaPipe + joint angle math + real-time feedback is genuinely complex and impressive.
Demo Factor	Judges see it work live on a phone. The green/red skeleton is an instant wow moment.
Scalability	Can expand to include AI-generated custom plans, teleconsultation integration, or wearable data.

"We bring the physiotherapist into every patient's living room."
